

Course Syllabus: Data Structures & Algorithms

Department: Computer Science | Term: Spring 2025 | Credits: 3
Instructor: Dr. A. Rahman

Course Objectives:
This course introduces fundamental data structures (arrays, linked lists, trees, graphs, hash tables) and algorithm design techniques (recursion, divide-and-conquer, dynamic programming), with emphasis on complexity analysis.

Weekly Schedule

Week 1-2	Arrays, Linked Lists, Complexity Analysis
Week 3-4	Stacks, Queues, Recursion
Week 5-6	Trees, Binary Search Trees, Heaps
Week 7-8	Graphs, Traversals, Shortest Paths
Week 9-10	Hash Tables, Dynamic Programming
Week 11-12	Sorting Algorithms, Project Work

Grading Breakdown

Assignments	20%
Midterm Exam	25%
Final Project	25%
Final Exam	30%